Early correctness signals for efficient code generation

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Research question and contribution

Can early hidden states distinguish successful and unsuccessful code-generation attempts for the same problem, improve on observable decoding statistics, and support economical interruption? These are separate claims. A predictor that recognizes easy prompts may have high pooled AUC while failing to distinguish attempts within a prompt. We therefore separate task difficulty, within-task discrimination, probabilistic prediction, and sequential operational value.

The contribution is a bounded, auditable empirical study and an explanatory analysis of a negative result. It does not claim to invent hidden-state probing, PCA, early abort, or geometric stopping-cost identities. The study examines when readout capacity and threshold transport obstruct practical use of predictive signals.

Design and validation

Qwen2.5-Coder-1.5B-Instruct generates stochastic Python candidates using immutable weights/tokenizer, temperature 1.0 selected by a separate 960-trajectory pilot, top-p 0.95 and a 512-token limit. The source bank contains 2,640 attempts across 132 HumanEval+ tasks; the frozen transfer bank contains 640 attempts across 32 MBPP+ tasks. Official EvalPlus 0.3.1 base and plus tests define functional success. The complete bank retains twenty attempts per task, including after early successes.

Five outer task-grouped folds evaluate held-out tasks. Three inner grouped folds select layer and L2 regularization using task-balanced log loss. Calibration scores cross-fit the entire selection procedure. At the fixed 32-token endpoint, eligibility is strictly $T>32$: 2,158 candidates remain, across 132 tasks, of which 92 contain both outcomes. Within-task AUC gives each mixed task equal weight; proper losses retain homogeneous tasks. Paired task bootstrap intervals condition on fitted scorers and do not capture all retraining uncertainty.

Original results

Predictor	Pooled AUC	Within-task AUC	Log loss
Observables B	0.701	0.628	0.630
Hidden H	0.592	0.568	0.687
H+B	0.646	0.581	0.668

The H+B minus B within-task AUC difference is −0.0470, 95% interval [−0.0994, 0.0056]. The log-loss difference is +0.0379 [0.0067, 0.0706]. The original incremental and joint operational criteria are not met. This delimits the tested procedure; it does not establish that hidden states contain no correctness information. Independent pair counting and reconstruction of all three primary scorers reproduce the archived probabilities to numerical precision.

Explanatory follow-up: capacity, decisions, and economics

All follow-up analyses were defined after the original results and remain exploratory. Observable ablations retain grouped internal selection. Mean NLL and margin alone preserve much of the within-task ranking, but their log loss is worse than the full observable baseline. PCA fitted exclusively within training partitions and an anchored residual readout reduce the original combined-model deficit. PCA64 achieves within-task AUC 0.630 and log loss 0.632; the residual achieves 0.635 and 0.629. Their paired intervals against B include zero. Stronger joint regularization alone does not resolve the deficit.

The residual uses logit(p_B) as a fixed offset and an L2-penalized hidden-state correction. Training offsets exclude their own labels from first-stage fitting, selection, and calibration; this separation is repeated inside residual selection. These controls support sensitivity to parametrization without isolating dimensionality as a unique cause or proving complete signal redundancy.

Frozen threshold transfer exposes a different failure mode. H+B retains only 276/366 active correct MBPP+ candidates: 75.41% [61.87%, 87.74%], despite achieving the 97% empirical development target. Ninety correct candidates are discarded across twelve tasks; five tasks account for sixty discards. Two tasks lose all their available successes. Rates among all potential candidates and initiated candidates differ because earlier decisions determine which later attempts start. Source OOF, source gate development, and full-source-fitted target scoring are reported separately; the comparison does not identify a pure causal domain effect.

An idealized label-aware oracle aborts incorrect active candidates and preserves every correct one on the same ordered stream. At eight tokens and N=20, its token savings against verifier stopping are 77.75% on HumanEval+ and 76.25% on MBPP+, with identical observed resolution. This measures conditional economic headroom, not an implementable predictor or a universal lower bound for lossy gates. With iid attempts conditional on a task, mean attempt cost d, success probability p and global correct retention r, expected cost is d times the truncated geometric sum with ratio $1-pr$. For an unlimited budget and positive p,r, lower expected cost requires $\frac{d}{E[T]} < r$. Within-attempt cost and success may depend. Task heterogeneity, finite budgets, coverage losses and gate overhead limit direct use of pooled formulas.

Limitations, next experiment, and research skills

The evidence concerns one small model and two function-level benchmarks. Twenty-six source tasks appeared in earlier local artifacts; researcher novelty and absence of pretraining contamination are not claimed. Execution was prospectively authorized by the student, while advisor approval remains pending; no independent preregistration is claimed. New analyses are post-hoc and their intervals are pointwise. Passing tests is operational correctness, not formal verification; the checker is not a complete security sandbox. Token savings are not latency, energy, or FLOPs savings.

A proposed follow-up reserves twenty disjoint technical tasks and 120 additional MBPP+ tasks, with twenty attempts each, to test the paired retention difference between source-frozen B and H+B gates at the same eight-token checkpoint. The single endpoint, IDs, inference, precision and resource budget are written before new outcomes. Collection awaits student approval and advisor review.

The project demonstrates task-grouped experimental design, nested model selection, strict cross-fitting, PyTorch instrumentation, verifier integration, paired statistical estimation, stopping-cost analysis, and reproducible software with hashes, regression tests and auditable artifacts. It provides concrete methods and limitations that can be examined in a research interview and thesis defense.

Selected context: Hewitt & Liang (2019), probing controls; Guo et al. (2017), calibration; Ovadia et al. (2019), uncertainty under shift; Ruan et al. (2026), independently certified early abort in agent episodes. Full references, source metadata, data provenance and commands accompany the thesis.